

Cortical motor activity modulates respiration and reduces apnoea in neonates

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eLife Assessment

Zandvoort and colleagues describe respiration-brain coupling in the context of apnoea in human newborns. The authors have addressed an **important** question and supported their claims with **solid** data. The rigor of the findings could perhaps be further strengthened with some relatively minor changes to the analysis methodology.

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Abstract

Respiration is governed by a widespread network of cortical and subcortical structures. This complex communication between the brain and lungs is altered in pathological conditions. Apnoea – the cessation of respiration – is a common condition in infants, particularly those born prematurely. Apnoea in infants is believed to relate to immaturity of brainstem respiratory centres; involvement of the cortex in respiration in infants has yet to be explored. We investigated if cortical control of respiration occurs in newborn humans and whether it relates to apnoea. Using simultaneous electroencephalography (EEG) and impedance pneumography, we show that cortico-respiratory coupling is present in premature and term newborns (104 recordings from 68 infants; 34.5 ± 2.6 weeks post-menstrual age), identifying an interplay between breathing phase and EEG amplitude. We further shed light on the biological meaning by revealing that the strongest coupling occurs during inspiration and that cortical activity precedes respiration, with coupling strongest over frontocentral regions. These findings support the notion that the cortico-respiratory coupling observed here primarily constitutes communication between cortical motor areas and lung effectors. Moreover, we show that cortico-respiratory coupling is negatively correlated with the rate of apnoea, revealing novel mechanistic insight into this common and potentially life-threatening neonatal pathology.

Introduction

Apnoea – the cessation of respiration – is a potentially life-threatening event¹, affecting more than 50% of preterm infants and some term infants². Apnoeic episodes can result in physiological instability including oxygen desaturation and bradycardia¹, are associated with suppression of brain activity³, and have been linked to poorer neurodevelopmental outcomes⁴. Apnoea of prematurity is caused by immaturity of the lungs, chest wall, and nervous system⁵. Immaturity of the brainstem plays a significant role in apnoea¹; however, the relationship with other respiratory-linked brain regions is less clear. In adults with congenital central hypoventilation syndrome⁶ or obstructive sleep apnoea⁷, a respiratory-linked increase in cortical motor activity suggests that the motor cortex plays an important role in maintaining autonomous respiration.

Recently, it was proposed that communication between the cortex and lungs can be identified and quantified through so-called cortico-respiratory coupling^{8,9}. This functional coupling arises when the amplitude of cortical activity is modulated by the respiration phase, or *vice versa*. This phase-amplitude coupling is typically quantified using non-invasive recordings capturing respiratory and neural activity (e.g., magneto- or electroencephalography [EEG]). Studies have demonstrated that the respiratory phase at low frequencies (<1 Hz) relates to higher-frequency oscillations in the cortex^{9,10}. Phase-amplitude coupling therefore allows us to quantify the interplay between respiratory and cortical activity. In adults, this coupling can be observed in a wide network of subcortical and cortical regions including motor areas⁹. Moreover, cortico-respiratory coupling is aberrated in pathological conditions¹¹.

Here, using cortico-respiratory coupling, we investigated whether cortical motor areas play a role in respiration in infants (**Figure 1**). We hypothesised that the strength of cortico-respiratory coupling is negatively associated with apnoea rate. To this end, we first confirmed the existence of cortico-respiratory coupling in both premature and term infants. To investigate if the coupling relates to the central motor drive, we further studied the spatial localisation, timing throughout the respiratory cycle and coupling directionality. Finally, we examined the relationship between cortico-respiratory coupling and apnoea rate.

Results and Discussion

Respiratory and cortical activity exhibit phase-amplitude coupling in infants

We analysed simultaneous 8-channel resting-state EEG (to measure cortical activity) and impedance pneumography (IP; to measure respiration) in 68 infants (28–42 weeks postmenstrual age [PMA] at time of recording) on 104 occasions. We first assessed whether and how EEG and IP are interrelated using cross-frequency coupling^{9,12,13}. In line with findings in adults⁹, we expected that the amplitude of oscillatory brain activity is modulated as a function of the respiratory cycle. That is, we expected to detect an EEG component that systematically varied with timing within the respiratory cycle. To this end, we identified individual breaths (and thus respiratory phase) from the IP signal¹⁴. Through averaging the EEG (time-locked to the IP signal) across breaths, we observed that EEG amplitude was modulated in the delta-(0.5–4 Hz) and theta-band (4–8 Hz) frequencies depending on the respiratory phase (**Figure 2A–B**). This analysis can be seen as a first indication that the EEG amplitude in specific frequency ranges changes systematically with the respiratory phase.

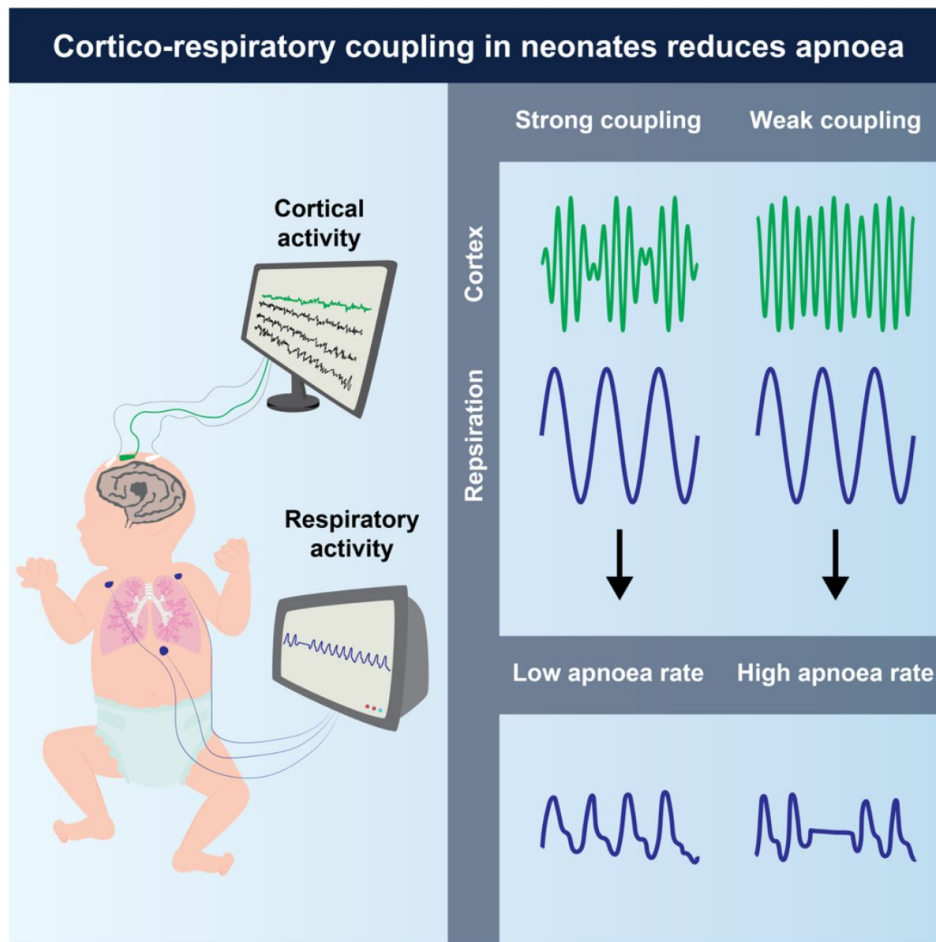


Figure 1.

Schematic of the data acquisition setup (left panel) and main findings (right panel).

Each recording consisted of cortical activity (using electroencephalography) and respiratory activity (using impedance pneumography).

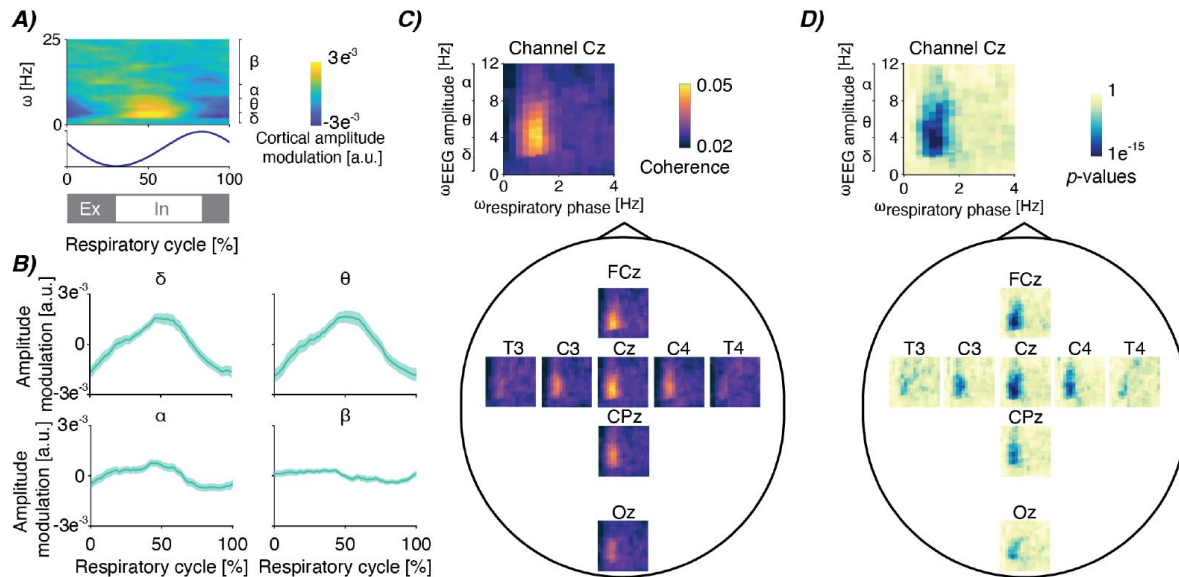


Figure 2.

Cortico-respiratory coupling in infants.

A) Electroencephalographic (EEG) amplitude modulation depending on the respiratory cycle and frequency. EEG amplitude is averaged over breaths, recordings and channels FCz and Cz. The blue graph below is the impedance pneumographic signal time-locked from breath to breath and averaged over breaths and recordings. Grey and white bars below indicate expiration (Ex; in grey) and inspiration (In; in white). **B)** The time-resolved amplitude modulations averaged within four frequency bands (delta: 0.5-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, beta: 13-25 Hz), focusing on channels FCz and Cz again. Shaded light-green area is the standard error across recordings after pooling over channels. **C)** Spatial map of coherence-based phase-amplitude coupling (PAC) between the respiratory phase ($\omega_{\text{respiratory phase}}$) and EEG amplitude ($\omega_{\text{EEG amplitude}}$) – see Figure S1 for the pan-spectral PAC estimates. Spectra are the mean over all recordings. **D)** Spatial map for the corresponding statistical significance relative to surrogate PAC obtained through epoch shuffling (reported on a logarithmic scale).

We then statistically confirmed this association between EEG and respiration using phase-amplitude coupling (PAC). Using cross-frequency coherence, we observed significant PAC between the phase at the mean respiratory frequency and the EEG amplitude in delta and theta frequency bands (**Figure 2C-D**). PAC was strongest at central electrodes (channels FCz and Cz; **Figure 2C-D**). In line with the results presented in **Figure 2A**, we did not find significant PAC at higher EEG frequencies such as the alpha (8-13 Hz) and beta (13-25 Hz) frequency bands (Figure S1).

These findings confirm that cortico-respiratory coupling occurs in infants. Whilst we are the first to show that cortico-respiratory coupling already exists in early life, adult studies have revealed widespread brain-respiratory networks covering both cortical and subcortical areas, as well as pan-spectral components of oscillatory brain activity (2-150 Hz)⁹. These networks have been linked to a diverse range of perceptual, cognitive and motor processes^{8,15-17}. In these studies, respiratory rhythm has been found to relate to task performance (e.g., better performances are found during a certain respiratory phase) as well as neural oscillations in the brain¹¹. For example, during an isometric motor task, the strength of cortico-muscular communication at beta-band frequencies cyclically changes with the respiratory phase¹⁶. Moreover, the probability of initiating goal-directed movements relates to cortical motor activity (here, measured as the readiness potential amplitude) and respiratory phase¹⁷. These relationships between task performance, respiration and cortical activity are well-established and replicated in a variety of animal and human adult studies, demonstrating the importance of cortico-respiratory coupling.

Cortico-respiratory coupling is strongest during inspiration and driven by the cortex

To further shed light on the functional meaning of cortico-respiratory coupling in neonates, we next examined the timing and directionality of the coupling. To study whether coupling strength fluctuates throughout the respiratory cycle, we focused on the EEG channels with strongest coupling (FCz and Cz), and frequencies where statistically significant coupling occurs (**Figure 2D**; alpha level set at 0.001). We confirmed that PAC changes in a time-dependent manner, observing the strongest PAC during the inspiratory phase (**Figure 3A-B**). Such within-breath modulation is evident for both delta- and theta-frequencies (**Figure 3A-B**). This indicates that the strength of cortico-respiratory coupling changes throughout the respiratory cycle. During the inspiratory phase, neural activation from the central nervous system is required to drive respiratory motoneurons causing contraction of respiratory muscles. Given the phasic PAC modulation as seen in **Figure 3A-B**, this suggests that cortical areas centred around the vertex play an important role in the neural activation of respiratory muscles.

Standard coherence (i.e., our PAC measure) only quantifies linear coupling in a bidirectional manner and cannot directly show any directionality, meaning that it is not possible to distinguish whether cortical activity precedes or follows respiratory activity. To unravel whether the cortical activity drives respiration or *vice versa*, we employed a cross-frequency version of the phase-slope index¹³. This revealed that delta- and theta-band EEG amplitude leads respiratory activity (**Figure 3C**), demonstrating that the cortical activity drives respiration. We observed the strongest phase-slope index around central areas, in particular for channel FCz (**Figure 3D**). Moreover, in line with the PAC analysis (**Figure 2**), phase-slope index was not statistically significant for alpha- and beta-band frequencies (Figure S2).

Considering the cortico-respiratory coupling's spatial location, its timing in the respiratory cycle and its directionality of action leads us to conclude that this coupling observed represents the respiratory drive from higher-order motor centres, primarily encompassing the cortical motor areas. Structurally, the motor cortex connects to respiratory muscles both directly and indirectly¹⁸. This allows for functional control over respiratory muscles (comprising diaphragm and intercostal muscles) via the corticospinal¹⁹ and reticulospinal tracts²⁰. In cat preparations, transection of the ventrolateral part of the spinal cord (which essentially blocks the

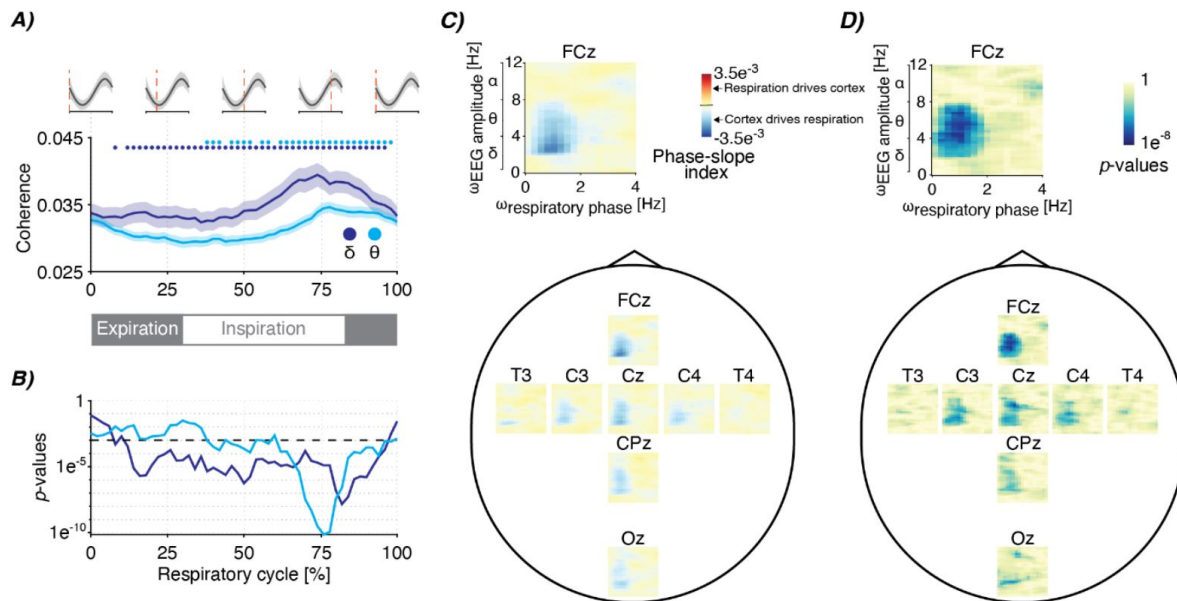


Figure 3.

Timing and directionality of the cortico-respiratory coupling.

A) Phase-amplitude coupling modulation (estimated as cross-frequency coherence) throughout the respiratory cycle. Graphs are colour-coded for the delta- and theta-frequency bands (dark- and light-blue colours respectively), which are the EEG's primary frequency bands for which we observed statistically significant PAC (Figure 2C-D). Shaded regions indicate standard error over recordings. PAC was estimated for EEG channels FCz and Cz. Grey waves at the top and grey bar at the bottom indicate the phase of the respiratory cycle. Coloured dots indicate points that are statistically significant (alpha level was set to 0.001). **B)** Corresponding statistical significance analogous to the coloured dots at the top of panel A (black dashed line indicates the alpha level). **C)** Spatial map of the cross-frequency phase-slope index between respiratory phase ($\omega_{\text{respiratory phase}}$) and EEG amplitude ($\omega_{\text{EEG amplitude}}$) (see Figure S2 for the pan-spectral estimates) and **D)** its statistical significance (reported on a logarithmic scale). The spectra show the mean over all recordings.

brainstem input to respiratory motoneurons) does not prevent cortical stimulation from eliciting responses in the respiratory intercostal muscle motoneurons, signifying the cortico-spinal drive²⁰. Such motor-evoked potentials can also be elicited by stimulating the cortical motor areas of humans using transcranial magnetic stimulation^{21,22}.

Whilst we believe that it is likely that our coupling reflects cortical motor drive, one might also argue that cortico-respiratory coupling could reflect changes in blood flow or oxygen saturation which induces ‘coupling’, where blood flow/oxygen saturation acts as a confounding factor essentially modulating respiratory and brain activity in a cycle-dependent manner. It is, however, unlikely that this is the case for our results given the focal spatial localisation around central electrodes and narrow spectral bandwidth comprising EEG frequencies between 1-8 Hz. Moreover, the amplitude modulation of blood flow reveals a different time-frequency representation over the respiratory cycle (Figure S3) when compared to cortical activity (**Figure 2A**).

To further our understanding of whether the coupling in infants relates directly to central motor drive, future work should examine potential coupling in the cortico-muscular domain. Electrical activity of the diaphragm (Edi) in infants can be recorded using an Edi catheter²³, as is used in ventilators with neurally adjusted ventilatory assist²⁴. Correlating this electromyographic (EMG) activity from the diaphragm with brain activity (e.g., EEG) will enable cortico-muscular assessments that can quantify whether oscillatory components in the EEG and diaphragmatic EMG are phase-locked. It is of particular interest whether linear coherence can be observed in the frequency range of 1-8 Hz, which is expected from our PAC findings presented in **Figure 2**. Interestingly, such coherence analysis has been conducted at the muscular level in rodents. Invasively recording activity from the diaphragm and intercostal muscles revealed strong inter-muscular coherence in the lower frequencies between 1 and 8 Hz²⁵. Future work should also establish to what extent the cortical motor areas provide control over other respiratory functionalities such as altering respiratory rate. In the current investigation, we focused on how cortical amplitude directly related to the respiratory phase but phase-amplitude coupling of cortical activity has also been revealed to play a role in adjusting respiratory rate²⁶ (i.e., respiratory rate rather than individual breaths).

Cortico-respiratory coupling relates to apnoea in infants

Having identified significant cortical-respiratory coupling in infants, we then tested our main hypothesis - that the strength of the cortico-respiratory coupling is negatively associated with apnoea. Apnoea of prematurity can lead to periods of physiological instability when apnoea is accompanied by bradycardia and desaturation⁵. These episodes may have long-term neurological effects⁴ and, hence, it is important to better understand the pathophysiology. In addition to pulmonary immaturity, apnoea of prematurity has been primarily attributed to immaturity and weakened neural drive from the respiratory centres in the brainstem. These respiratory centres are principally located in the medulla oblongata and pons within the brainstem, which ensure the periodical activation of lung muscles to enable respiration. However, as our results indicate, further central input to respiration is provided by higher-order brain areas such as the cortical motor areas.

To test our hypothesis, apnoea rate was estimated over a time window of up to 24 hours (infants have their vital signs continuously monitored at the neonatal care unit), whereas cortical-respiratory coupling was assessed on the same day from EEG recordings with an average duration of 1.7 ± 0.6 hours. Choosing different time windows ensured that any relationship between apnoea rate and cortical-respiratory coupling was not the mere result of the number of breaths included in the analysis. For instance, when the same respiratory signal is used in both analyses, a higher apnoea rate would mean that fewer windows are used for the PAC estimation which could result in an increased magnitude. We selected PAC pooled over EEG channels Cz and FCz since their coupling magnitudes were the highest.

We found support for our hypothesis that apnoea and cortico-respiratory-coupling are related. PAC was negatively associated with apnoea rate ($t(98):-2.28$, $p:0.02$, $\pi:-0.22$; **Figure 4A**); this relationship was also identified when focusing on individual channels [Figure S4] and preterm infants younger than 36 weeks [Figure S5]). This suggests that a stronger cortico-respiratory coupling may reduce the likelihood of experiencing apnoeas in infants. This negative association provides further support that our measures of cortico-respiratory coupling (**Figures 2-3**) are related to central motor drive. In this case, cortical motor activity assists the brainstem's central pattern generators (CPGs) to establish increased control over the respiratory muscles, which in our analysis, is reflected as an association with apnoea rate. The motor cortex can exert coordination over the CPGs through its downstream projections to the thalamus²⁷, which, in turn is connected to the preBötzing complex (the structure that primarily hosts the CPGs²⁸).

In theory, the strength of cortical-respiratory coupling and its inverse relationship with apnoeas could reflect a shared relationship with PMA, where both coupling strength and apnoea rate change depending on PMA. The apnoea rate is expected to decrease with increased PMA across a broad age range²⁹. It is also plausible that the strength of cortico-respiratory coupling (i.e., PAC magnitude) may change with PMA, with younger (i.e., preterm) infants potentially having weaker coupling. However, in our dataset, we observe statistically significant PAC in preterm infants ($33^{+0}-36^{+6}$ weeks^{+days}; **Figure 4B**), and very preterm infants ($28^{+0}-32^{+6}$ weeks^{+days}; **Figure 4C**). Moreover, PAC and apnoea rate were not significantly associated with PMA (Figure S6A-B, linear mixed-effects analysis). Similarly, PAC and apnoea rate were not significantly related to postnatal age (PNA; defined as the age in weeks from birth; Figure S6C-D). Whilst future studies should investigate the relationship between cortico-respiratory coupling and age across an even broader age range, the lack of significant associations here suggests that the relationship we observed between apnoea rate and cortico-respiratory coupling is not determined by age.

It is known that most apnoeas in infants occur during active sleep^{6,30} and delta- and theta-band frequencies in EEG are strongly related to sleep state³¹. We speculate that cortico-respiratory coupling may change depending on sleep state, giving rise to more episodes of apnoea during certain sleep stages. Indeed, adult studies have suggested that cortical respiratory drive may occur predominantly during wakefulness^{6,32}. Future work should study the effects of sleep stages on cortico-respiratory coupling to shed further light on the role of cortical motor drive for respiration in infants.

In summary, we demonstrate the existence of cortico-respiratory coupling in newborn infants. We believe that this coupling is related to central motor drive due to its spatial location (predominantly over frontocentral EEG channels), timing (during inspiration) and directionality (with the cortex driving respiration). Consistent with our hypothesis, we found a significant relationship between apnoea rate and coupling strength, with infants with stronger cortico-respiratory coupling less likely to experience apnoea. This has far-reaching implications for our mechanistic understanding of apnoea of prematurity as it demonstrates that cortical motor areas play a pivotal role in apnoea in infants.

Material & Methods

Participants and study design

This study was conducted at the Newborn Care Unit of the John Radcliffe Hospital, Oxford University Hospitals NHS Foundation Trust, Oxford, UK. The National Research Ethics Service approved the study (references: 19/LO/1085; 12/SC/0447). Eligible families were approached and given verbal and written details about the study. Parents provided written consent to take part in the study. The study conformed to the standards set by the Declaration of Helsinki and Good Clinical Practice.

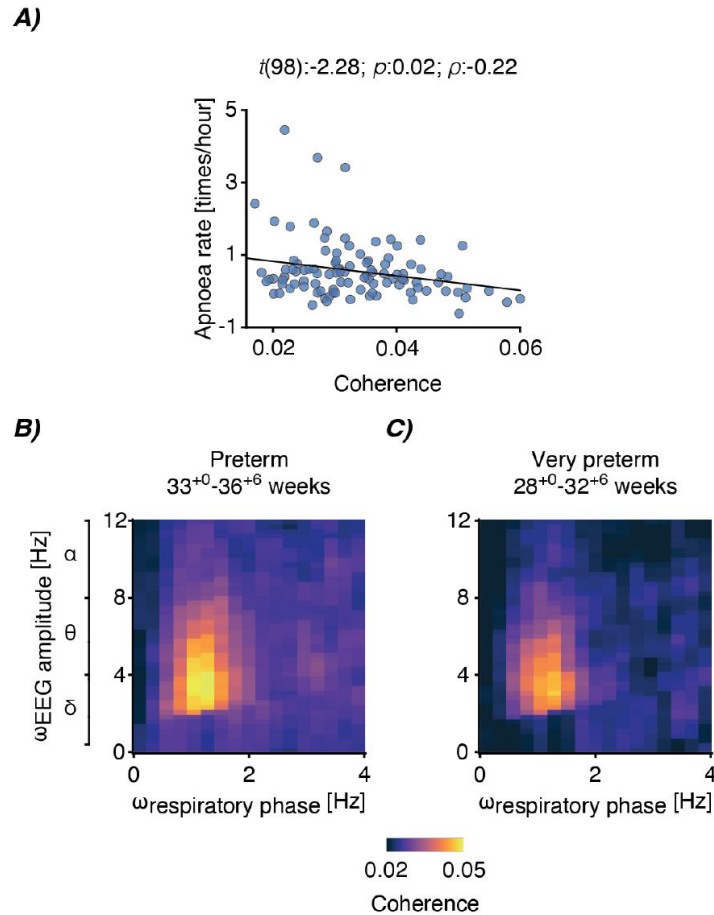


Figure 4.

Cortico-respiratory coupling dependency on apnoea rate and postmenstrual age.

A) Relationship between apnoea rate and statistically significant phase-amplitude coupling (PAC; defined as coherence encompassing delta- and theta-band frequencies). Each blue dot is the data of an individual recording (data of 68 infants were included from 104 test occasions). Black linear graph is the best fit of the linear mixed-effects model (fixed factors: coupling averaged over statistically significant samples from channels FCz and Cz [see Figure S4 for channel-specific relationships], data length to determine apnoea rate, mode of ventilation, and postmenstrual age; random factor: infant). Panel title contains the test statistic of the predictor's regression slope (t), its significance (p), and partial correlation coefficient (ρ). **B-C)** PAC spectra between the respiratory phase ($\omega_{\text{respiratory phase}}$) and EEG amplitude ($\omega_{\text{EEG amplitude}}$) for **B)** preterm (58 recordings; 33.0-36.86 weeks), and **C)** very preterm (28 recordings; 28.0-32.86 weeks) infants. Both spectra are averaged over recordings and EEG channels FCz and Cz.

Clinically stable infants aged between 28- and 42-weeks postmenstrual age (PMA) at the time of the test occasion were eligible to be included in the study. Infants were excluded if they had intraventricular haemorrhage grades III or IV, hypoxic ischaemic encephalopathy, congenital malformations, there was a known history of maternal substance misuse during pregnancy, or at the time of the study, they were receiving mechanical ventilation, opioid analgesics, or antibiotics (as a surrogate marker of infection). These conditions are known to alter respiratory and brain activity. We enrolled 74 infants who were recorded on 123 separate occasions (range: 1-8 recordings per infant). Fifty-one infants were included in one or two test occasions; 23 infants took part in the ongoing *Breathing and Brain Development* study where they are studied approximately once a week whilst in the Newborn Care Unit (included on 3 ± 2 [median $\pm$ interquartile range] occasions). Demographic details of all infants can be found in **Table 1**.³³ Parts of this dataset have been reported earlier in Zandvoort et al.³³ to address a different research question.

Data recordings

During each test occasion, approximately one to two hours of simultaneous electroencephalography (EEG) and vital signs (respiration, heart rate and oxygen saturation) were recorded. We continuously recorded the vital signs throughout their hospital stay of the infants who took part in the *Breathing and Brain Development* study.

Vital signs were recorded for each infant using Phillips IntelliVue MX800 or MX750 monitors. These data were continuously downloaded from the vital signs monitor using an electronic data capture software (iXtrend, iXitos, Germany). Both electrocardiograph (ECG; sampling frequency: 250 Hz) and impedance pneumography (IP; sampling frequency: 62.5 Hz) were recorded with three electrodes placed on the infant's chest. Peripheral photoplethysmography was recorded at 125 Hz from a probe placed on the infant's foot or hand.

EEG was recorded from DC to 400 Hz using a SynAmps RT 64-channel headbox and amplifiers and CURRYscan7 neuroimaging suite (Compumedics Neuroscan) at a sampling rate of 2 kHz. Eight electrodes were placed at Cz, CPz, C3, C4, Oz, FCz, T3, and T4. The ground electrode was placed at FPz and the reference electrode at Fz. The infant's scalp was cleaned with NuPrep gel (D.O. Weaver and Co., Aurora, USA) to achieve impedances of 5-10 k Ω . A conductive paste (Elefix EEG paste, Nihon Kohden, Tokyo, Japan) was used to attach disposable Ag/AgCl cup EEG electrodes (Ambu Neuroline, Ballerup, Denmark) to the infant's scalp. EEG was recorded for an average of 1.7 ± 0.7 hours (mean $\pm$ SD). ECG was co-registered with our EEG recordings using a single electrode (Ambu Neuroline, Ballerup, Denmark) placed on the infant's chest (referenced to Fz).

To synchronise the vital signs and EEG recordings, we registered approximately 30 annotations simultaneously on both recordings using a bespoke triggering device (the PiNe box)³⁴. Annotations were used to mark resting state as well as stimulus-evoked activity (these responses were used in other studies to investigate infant brain maturity – see Zandvoort et al.³³ – but were used here only to synchronise the EEG and vital signs recordings). The PiNe box enables synchronisation with an average precision of 105 ± 14 ms (mean $\pm$ SD)³⁴. Annotation timings were matched between both vital signs and EEG recordings. Next, to further improve synchronisation, we utilised the ECG signals which were recorded on both the vital signs and EEG recording devices. To do so, ECG epochs were created from -10 to 10 seconds around each annotation. We aligned the ECG signals around each of these annotation epochs using cross correlations, providing a temporal delay between vital signs and EEG.³⁴ Temporal synchronisation between vital signs and EEG recordings was achieved by taking the median of the time differences across all annotations.

From the total of 123 recordings, we excluded 19 recordings due to (a) missing IP recordings (5 recordings), (b) missing annotations (9 recordings), and (c) inaccuracies in EEG-IP syncing (5 recordings; standard deviations over annotations exceeding 1 sec for annotation alignment or 0.1

Factors	
Age	
Postmenstrual age at recording (weeks)	34.5 ± 2.6
Gestational age at birth (weeks)	31.1 ± 3.4
Postnatal age at recording (weeks)	3.4 ± 2.2
Birthweight (g)	1,726 ± 816

Table 1.

Infant demographics.

Reported values are mean and standard deviation or number of babies or recordings. Values are reported over recordings for ages (postmenstrual, gestational, and postnatal), respiratory distress syndrome, persistent pulmonary hypertension, infection and respiratory support. Birthweight, sex, mode of delivery, and Apgar scores are provided over infants.

Sex	
Females	27
Males	47
Mode of delivery	
Normal vaginal delivery	16
Vaginal assisted (ventouse/forceps/kiwi)	4
Elective Caesarean section	7
Emergency Caesarean section	47
Apgar scores	
Apgar at 1 minute	7.2 ± 2.2
Apgar at 5 minutes	9.1 ± 1.4
Apgar at 10 minutes	9.7 ± 0.8
Respiratory distress syndrome	
Ongoing condition	39
Past treated	48
No	36
Persistent pulmonary hypertension	
Past treated	5
No	118
Infection	
None	48
Treated	75
Respiratory support	
High flow therapy	25
Low flow therapy	16
Spontaneous ventilation	82
Caffeine received on the day of the EEG recording	
Yes	59
No	64

Table 1. (continued)

sec during ECG alignment). Hence, data comprising 104 recordings from 68 infants were used in the analysis.

Breath and apnoea identification

IP signals were processed to identify breaths (including apnoeas). To this end, we used an algorithm validated to detect inter-breath intervals in infants¹⁴. Briefly, the algorithm attenuates noise introduced by movements and cardiac activity by filtering the signal. An adaptive amplitude threshold then identifies individual breaths. This threshold was set at 0.4 times the standard deviation of the signal over the preceding 15 breaths. Finally, a support vector machine algorithm was applied to all long (here, 15 seconds or greater) inter-breath intervals to separate episodes of apnoea (cessation of respiration) and periods of low amplitude erroneously identified as apnoeas due to noise or shallow breathing. The inter-breath intervals marked as noise/shallow breathing were discarded from further analysis. The model was trained and validated on IP signals acquired during a clinical trial³⁵, which was a completely independent data sample to the one used in the current study.

Phase-amplitude coupling

In line with Osipova *et al.*¹², we defined phase-amplitude coupling (PAC) as cross-frequency coherence between a low-frequency phase and high-frequency amplitude. Note that the term “phase-amplitude coupling” is used interchangeably in the literature to mean the coupling principle of the statistical relationship between phases and amplitudes as well as the actual measure to detect this. We reserve the abbreviation for the latter. For our PAC measure, this meant that coherence was computed between the filtered IP signal and instantaneous amplitude of the EEG signal at some higher frequency (i.e., the second-order moment between the IP phase and EEG amplitude). More formally, in our case PAC was defined as:

$$C(f_1, f_2) = \frac{|\sum_{n=1}^N X(f_1) * Y(f_2)^*|}{\sqrt{\sum_{n=1}^N |X(f_1)|^2 * \sum_{n=1}^N |Y(f_2)|^2}} \quad \text{Eq. 1}$$

with X being the IP signal at frequency f_1 , Y the instantaneous EEG amplitude, and $*$ the complex conjugate. This instantaneous amplitude was obtained using the magnitude of the Hilbert transform after narrowly filtering the data at frequency f_2 . This component is often filtered with a bandwidth that relates to the low-frequency component f_1 . Following earlier recommendations, the filter bandwidth of the high-frequency component (at f_2) was equal to a single time the frequency of the low-frequency component f_1 ³⁶ (e.g., when f_1 equals 5 Hz the filter bandwidth of f_2 is set to 5 Hz).

Data were segmented into 4-sec windows (50% overlap), which meant that the frequency resolution was equal to 0.25 Hz (allowing to capture low delta-band frequencies), which comprised around four breaths. To focus on regular respiration, we discarded epochs which contained fewer than three detected breaths. An epoch was also removed from further analysis when its standard deviation exceeded 3 times the mean or median of the standard deviation of all epochs (which was done separately for the EEG and IP). Finally, we rejected any epoch containing an absolute value greater than 10 times the mean or median of the standard deviation of all epochs. Before cross-spectrum computation, EEG and IP epochs were convolved with a wavelet constructed from a Hanning window.

Changes in PAC throughout the respiratory cycle

To investigate how cortico-respiratory coupling changes throughout the respiratory cycle, we estimated PAC at 50 equally spaced timepoints within the respiratory cycle.

To this end, we time-locked both EEG and IP to each inter-breath interval as identified by the algorithm of Adjei *et al.* ¹⁴. At each respiratory cycle phase, cross- and auto-spectra were obtained to estimate PAC over all breaths within a recording. We limited the analysis to EEG channels FCz and Cz, because they exhibited the highest PAC magnitude (**Figure 2**), as well as statistically significant frequency-frequency samples relative to the surrogate PAC (see **Figure 2D** for the *p*-values of this statistical comparison). PAC was computed within the different frequency bands (delta: 0.5-4 Hz; and theta: 4-8 Hz) to see whether its modulation was specific for each frequency band. The same preprocessing steps apply as for PAC computation described earlier apart from the fact that data were segmented into 2/3 sec windows (so that temporal changes would appear within the respiratory cycle).

Directionality of phase-amplitude coupling

To assess whether the cortical activity was preceding or trailing relative to the respiratory activity, we employed a cross-frequency version of the phase-slope index. Briefly, this metric estimates the phase lags over narrow frequency ranges and quantifies whether they increase or decrease over this range by estimating the phase-frequency slopes. Increasing or decreasing slopes signify changing time lags depending on the frequency of the oscillation and can therefore be indicative of whether a signal leads or trails another signal. We used the phase-slope index (Ψ) as proposed by Jiang *et al.* ¹³, which was defined as:

$$\Psi(f_1, f_2) = \text{imag} \left(\sum_{f_2 - \frac{\beta}{2}}^{f_2 + \frac{\beta}{2}} \tilde{C}^*(f_1, f_2) \tilde{C}(f_1, f_2 + \Delta f) \right) \quad \text{Eq. 2}$$

with $\tilde{C}$ being the coherency (i.e., the normalised cross-spectrum) of C defined in Eq. 1. Δf is the frequency resolution and β the bandwidth used to estimate the phase slope. In line with previous recommendations by Jiang *et al.* ¹³, we set β to 1 Hz (i.e., four times the frequency resolution).

Surrogate analysis

To assess the statistical significance of PAC (i.e., coherence) and phase-slope index at each of the EEG channels, we performed a surrogate analysis under the null hypothesis that the means of the true and surrogate PAC (and phase-slope index) were equal. The surrogate analysis consisted of randomly shuffling epochs of the EEG amplitude after which PAC was computed with the IP signal. The random shuffling of the EEG amplitude epochs destroys any temporal relationships whilst retaining the signals' characteristics. We obtained a surrogate spectrum for each recording. Next, the true and surrogate spectra were statistically compared on a sample-by-sample basis using dependent *t*-tests ³⁷ (using the FieldTrip toolbox – version 20210325). This comparison was performed over all recordings. Statistical significance was defined using the false discovery rate correction (with an alpha level set at 0.001).

Statistical analysis of relationships between cortico-respiratory coupling and apnoea

We tested our main hypothesis that PAC is negatively associated with apnoea rate. Vital signs were continuously monitored for the infants that took part in the *Breathing and Brain Development* study, which allowed us to analyse their vital signs in longer time windows than the EEG. We here selected up to 24 hours of vital signs from the day of the EEG recording (mean $\pm$ standard deviation: 9.5 ± 9.9 hours) and expressed this as a rate per hour. We then constructed linear mixed-effects models to study the relationships between PAC and apnoea rate. We included the coherence magnitude as a fixed factor, which was averaged over the statistically significant samples from channels FCz and Cz (see Figure S4 for the channel-specific relationships). As fixed effects, we included (1) data length of the respiratory signal, (2) mode of ventilation ('Self-ventilating', 'Low flow', 'High flow'; Figure S7), and (3) PMA (Figure S6). Infant ID was included as a random effect. Data length, mode of ventilation and PMA were included as confounding factors.

P-values (for the regression coefficient of PAC) and partial correlation coefficients were reported from these models. The normality assumption was verified with Q-Q plots. We also tested the linearity assumption (see Supplementary text). For visualisation (**Figure 4** [↗](#)), we used the `plotAdjustedResponse` function in MATLAB to adjust for the fixed factors (calculating a separate model without the random effect of an infant). Regression lines are visualised by calculating the line-of-best-fit on the adjusted data.

Data and code availability

- Due to ethical restrictions, the raw data that support the findings of this study are available on reasonable request. Requests can be made directly to the corresponding author (caroline.hartley@paediatrics.ox.ac.uk) or directed to the Paediatric Neuroimaging Group (University of Oxford) through the institutional email: neonatal.research@ouh.nhs.uk. Researchers will be required to agree to destroy the data within the time limits stipulated in the ethically approved study protocols. Upon agreement, data will be provided within 1 month.
- Summary-level data (before regression model adjustments of **Figure 4** [↗](#)), and channel- and recording-specific PAC spectra are provided at https://gitlab.com/paediatric_neuroimaging/cortico_respiratory_coupling [↗](#).
- The codes for all analyses are available at: https://gitlab.com/paediatric_neuroimaging/cortico_respiratory_coupling [↗](#). The algorithm for inter-breath interval detection is online at: https://gitlab.com/paediatric_neuroimaging/identify_ibi_from_ip [↗](#).
- Any additional information required to reanalyse the data reported in this paper is available from the lead contact upon request.

Acknowledgements

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Additional information

Author contributions

Conceptualization: CSZ and CH; **Data curation:** FU, SR, OF, and CH; **Formal analysis:** CSZ; **Funding acquisition:** CH; **Investigation:** FU, SR, and OF; **Methodology:** CSZ; **Project administration:** CH; **Software:** CSZ; **Resources:** EA, KTSP and SFF; **Supervision:** CH and EA; **Visualization:** CSZ; **Writing – original draft:** CSZ; **Writing – review & editing:** all authors

Additional files

Supplementary material [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors investigated the extent to which phase-amplitude coupling (PAC) of respiratory and electrophysiological brain activity recordings was related to episodes of life-threatening apnoea in human newborns.

Strengths:

I want to commend the authors for acquiring unique and illuminating data; the difficulty in recording and handling these data has to be appreciated. As far as I can tell, Zandvoort and colleagues are the first to provide robust evidence for respiration-brain coupling in newborns. Their creative use of the phase-slope index for peripheral-central interactions is innovative and credible. If proven to be robust, the authors' findings have important implications well beyond the field of brain-body research.

Weaknesses:

While the analyses were overall competently conducted and well-justified, I was not entirely convinced by a few methodological choices, specifically i) the computation of PAC surrogates,

ii) details of the linear mixed-effects model, and iii) the electrode selection for linking phase-amplitude coupling to apnoea frequency.

<https://doi.org/10.7554/eLife.107081.1.sa3>

Reviewer #2 (Public review):

Summary:

The author's central hypothesis was that the strength of cortico-respiratory coupling in infants is negatively associated with apnoea rate. To prove this, they first investigated the existence of cortico-respiratory coupling in premature and term-born infants, the spatial localisation of the cortical activity and its relationship with the phase of the respiratory cycle, and the directionality of coupling.

Strengths:

The researchers used synchronised EEG and impedance pneumography to detect the phase amplitude coupling.

They have studied a wide range of gestations, from 28 weeks to 42 weeks, including males and females. Their exclusion criteria ensured that healthy babies were studied and potential confounders of impaired respiratory activity were avoided. Their sequential approach in addressing the objectives was appropriate.

Weaknesses:

As a neonatal clinician and neuroscientist, I have commented based on my expertise. I have not commented on signal processing.

I did not identify any major weaknesses in the study. Some minor weaknesses include:

- (1) Data relating to the cortical oscillations and the respiratory phase is given. However, whether this would lead to their hypothesis that the strength of cortico-respiratory coupling is negatively associated with apnoea rate is unclear. What preceding data enabled the authors to link the strength of coupling to the rate of apnoea?
- (2) If we did not know of data showing the existence of cortico-respiratory coupling in newborn infants, then should it not be the first research question to examine?
- (3) What are the characteristics of the infants who contributed data to establish the cortico-respiratory coupling (Figures 2 and 3)?
- (4) Although it is the most plausible direction of the relationship, with neural activation driving respiratory muscle contraction, how can the authors prove this with their data? Given that they show coherence between signals, how do we know that the cortical signal precedes the respiratory muscle contraction?
- (5) Apgar score is an ordinal variable. The authors should summarise this as median (range).

<https://doi.org/10.7554/eLife.107081.1.sa2>

Reviewer #3 (Public review):

Summary:

This is a strong and important report that presents a framework for understanding cortical contributions to neonatal respiration. Overall, the authors successfully achieved their goal of linking cortical activity to respiratory drive. Despite the correlational nature of this study, it is a crucial step in establishing a foundation for future work to elucidate the interaction between cortical activity and breathing.

Strengths:

(1) The introduction and use of workflows that establish correlational relationships between breathing and brain activity.

(2) The execution of these workflows in human neonates.

Weaknesses:

Interpretations related to causal inference, confounds of sleep and caffeine, and the spatial interpretation of EEG data need to be addressed to ensure that the data appropriately support the conclusions.

<https://doi.org/10.7554/eLife.107081.1.sa1>

Author response:

We would like to thank the reviewers for their helpful comments and critique of our manuscript. We plan to make the following revisions, which will improve the clarity of our manuscript and the robustness of our findings.

We will revise methodological details and interpretation throughout the manuscript. In particular, we will consider alternative methods for calculating surrogates. We intend to investigate the relationship between apnoea rate and phase-amplitude coupling at other electrodes as suggested by Reviewer 1, and we will revise the details of the linear-mixed effects models.

In relation to the comments raised by both Reviewers 2 and 3, we will carefully address the wording throughout the manuscript, including addressing the order of hypotheses, our interpretation of the directionality of the relationship between cortical and respiratory activity, and the connection between cortical-respiratory coupling and apnoea. We will further clarify the limitations of our recording setup and approach, in particular the limited EEG montage, and add further details with regards to sleep state and caffeine.

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